Supplementary Material

Progressive Reasoning Collapse in LLM-Based Generative Recommendation

This document supplements the main paper with extended experimental results, architecture details, hyperparameter sensitivity analyses, per-dataset ablations, and a reproducibility statement. Section references (e.g., Theorem 1, Table 3) refer to the main paper unless otherwise stated.

1 Scalability to Pre-Trained LLM Architectures

The main paper (Experiment 7) defers T5-Large (770M parameters) due to peak memory exceeding 16 GB VRAM on a single T4 GPU when combined with full-catalogue ranking. Here we present results for **T5-base** (250M parameters, 12-layer encoder–decoder, $d = 768$), fine-tuned for sequential recommendation on ML-1M and Amazon Beauty. This experiment has three objectives: (i) confirm that Progressive Reasoning Collapse manifests in a pretrained encoder–decoder backbone; (ii) demonstrate that CARR is architecture-agnostic; and (iii) assess whether pretrained representations affect SGI behaviour on the sparse Beauty dataset.

1.1 Fine-Tuning Protocol

We initialise the transformer encoder from public T5-base weights (`t5-base`, HuggingFace) and attach a learned item-ID projection head $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{d\times|\mathcal{I}|}$. The decoder is discarded; only the 12-layer encoder is used for sequence encoding, consistent with CARR's scratch-trained architecture. The CARR compression module is applied to encoder hidden states at layer k^*. Fine-tuning runs for 30 epochs (early stopping patience 5) using the same composite objective as the main experiments ($\mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{collapse}} + \mathcal{L}_{\text{evidence}} + \mathcal{L}_{\text{compress}}$). Peak GPU memory is 14.8 GB, within the T4 budget.

1.2 Results on ML-1M and Amazon Beauty

Table 1: T5-base Fine-Tuned Results on ML-1M, Compared to Scratch-Trained CARR

Method	HR@10	NDCG@10	$\mathcal{R}(12)$	$\widehat{S}_L$
Full-T5-base	0.0412	0.0198	3.413	0.0241
Fixed-Mid-T5 ($k = 6$)	0.0038	0.0017	2.791	0.0029
Fixed-Late-T5 ($k = 9$)	0.0031	0.0015	3.102	0.0031
CARR-T5	**0.0441**	**0.0213**	**3.681**	**0.0239**
Scratch-trained CARR (main paper)	0.0325	0.0140	3.526	0.0223

Key findings:

1. **PRC persists in pretrained backbones.** Fixed-Mid-T5 on ML-1M achieves $\mathcal{R}(12)$=2.791, an 18.2% drop from Full-T5-base (3.413), confirming the collapse phenomenon is model-family–agnostic and is driven by structural compression, not random-initialisation artefacts.

Table 2: T5-base Fine-Tuned Results on Amazon Beauty

Method	HR@10	NDCG@10	$\mathcal{R}(12)$	$\widehat{S}_L$
Full-T5-base	0.0083	0.0042	3.187	0.0308
Fixed-Mid-T5 ($k = 6$)	0.0131	0.0067	2.634	0.0031
CARR-T5+SGI	**0.0174**	**0.0089**	**3.402**	**0.2891**
Scratch-trained CARR+SGI (main paper)	0.0131	0.0067	3.011	0.3452

Table 3: FLOPs and Latency: T5-base with and without CARR (ML-1M)

Configuration	FLOPs (B)	Reduction	HR@10
Full-T5-base (12 layers)	48.3	—	0.0412
Fixed-Mid-T5 ($k = 6$)	29.1	39.8%	0.0038
Fixed-Late-T5 ($k = 9$)	38.8	19.7%	0.0031
CARR-T5 (adaptive $k^* = 9$)	42.7	11.6%	**0.0441**

2. **CARR transfers directly to T5-base.** CARR-T5 achieves HR@10= 0.0441 (+7.0% over Full-T5-base, +35.7% over scratch-trained CARR) and $\mathcal{R}(12)$=3.681, surpassing the uncompressed baseline in both metrics. This mirrors the main-paper pattern and confirms that collapse-regularised training generalises across backbone choices.

3. **Pretrained features amplify Beauty gains.** CARR-T5+SGI achieves HR@10= 0.0174 on Beauty, a +32.8% gain over scratch-trained CARR+SGI (0.0131), consistent with T5's semantic representations providing richer initial geometry for the collapse monitor to preserve.

4. **CARR selects $k^* = 9$ on T5-base**, yielding an 11.6% FLOPs reduction with a quality *improvement*. Fixed-Mid achieves the largest reduction (39.8%) at catastrophic quality cost (HR@10 0.0412→0.0038), reinforcing Theorem 3: operating at $k < k^*$ trades reasoning geometry for throughput.

2 Extended Recommendation Quality Results

Tables 4–8 report HR@5, HR@10, HR@20, NDCG@5, NDCG@10, NDCG@20, and MRR across all five datasets and all model groups. HR@10 and NDCG@10 match the values reported in the main paper (Table 3); the additional cut-off metrics and MRR are reported here for completeness. All evaluations use the full-catalogue ranking protocol without negative sampling.

3 Full Layerwise Collapse Metrics Across Datasets

The main paper (Table 1) reports $\mathcal{R}(l)$ at layers $\{3, 6, 9, 12\}$ for ML-1M only, which is sufficient to validate Theorem 1 on the primary benchmark. Here we reproduce the same analysis on Amazon Beauty and Amazon Toys using the archival checkpoint evaluations. The monotonic post-compression decay predicted by Theorem 1 is confirmed on both datasets.
Observations. On both Beauty and Toys:

- Every fixed-compression method exhibits a cliff-drop at its compression boundary, consistent with Theorem 1 on datasets beyond ML-1M.
- CARR maintains the highest $\mathcal{R}(12)$ among all compression methods, confirming that collapse-regularised training generically preserves multi-intent geometry irrespective of domain.

Table 4: Full Recommendation Metrics on ML-1M

Method	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20	MRR
Group I: LLM compression methods							
Full-LLM	0.0118	0.0300	0.0718	0.0062	0.0144	0.0219	0.0094
Fixed-Early	0.0013	0.0033	0.0058	0.0006	0.0014	0.0020	0.0024
Fixed-Mid	0.0011	0.0028	0.0046	0.0008	0.0014	0.0019	0.0025
Fixed-Late	0.0010	0.0026	0.0050	0.0009	0.0015	0.0020	0.0027
KV-Pruning	0.0122	0.0323	0.0721	0.0062	0.0138	0.0214	0.0093
Token-Pruning	0.0124	0.0316	0.0716	0.0064	0.0146	0.0221	0.0096
CARR	**0.0125**	**0.0325**	**0.0732**	0.0060	0.0140	0.0220	**0.0097**
Group II: LLM-based recommendation baselines							
LLMRec	0.0381	0.0783	0.1524	0.0218	0.0460	0.0714	0.0351
UniSRec	0.0112	0.0278	0.0671	0.0053	0.0129	0.0200	0.0085
Group III: Non-generative sequential reference							
SASRec	0.0013	0.0033	0.0061	0.0006	0.0015	0.0021	0.0024
BERT4Rec	0.0421	0.0866	0.1511	0.0241	0.0486	0.0723	0.0368
GRU4Rec	0.0150	0.0311	0.0623	0.0084	0.0164	0.0252	0.0126

Table 5: Full Recommendation Metrics on Amazon Beauty

Method	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20	MRR
Group I: LLM compression methods							
Full-LLM	0.0025	0.0063	0.0135	0.0014	0.0031	0.0049	0.0023
Fixed-Early	0.0044	0.0108	0.0175	0.0026	0.0058	0.0083	0.0046
Fixed-Mid	0.0040	0.0100	0.0200	0.0023	0.0056	0.0091	0.0043
Fixed-Late	0.0034	0.0084	0.0168	0.0020	0.0043	0.0070	0.0036
KV-Pruning	0.0049	**0.0119**	0.0233	**0.0028**	**0.0061**	0.0097	**0.0050**
Token-Pruning	0.0032	0.0077	0.0158	0.0017	0.0039	0.0064	0.0031
CARR+SGI	**0.0053**	**0.0131**	**0.0241**	**0.0028**	**0.0067**	**0.0109**	**0.0052**
Group II							
LLMRec	0.0048	0.0111	0.0221	0.0033	0.0068	0.0108	0.0054
UniSRec	0.0024	0.0059	0.0123	0.0012	0.0023	0.0038	0.0019
Group III							
SASRec	0.0004	0.0011	0.0014	0.0003	0.0005	0.0006	0.0004
BERT4Rec	0.0062	0.0159	0.0302	0.0040	0.0088	0.0141	0.0068
GRU4Rec	0.0034	0.0084	0.0156	0.0018	0.0035	0.0056	0.0028

- The sharpest single-layer drop for Fixed-Mid occurs at layer 7 (the first post-compression layer), not at the boundary layer itself, consistent with the theoretical model: the compression at layer k acts on the output of that layer, so the effect first manifests in the subsequent layer's $\mathcal{R}$ value.
- Amazon Toys shows a smaller pre-compression $\mathcal{R}$ range (2.87–3.72) than ML-1M (2.84–5.00), reflecting sparser interaction structure; nevertheless, the post-compression ordering is preserved.

Table 6: Full Recommendation Metrics on Amazon Toys

Method	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20	MRR
Group I: LLM compression methods							
Full-LLM	0.0033	0.0088	0.0182	0.0019	0.0049	0.0079	0.0036
Fixed-Early	0.0040	0.0103	0.0193	0.0021	0.0050	0.0081	0.0040
Fixed-Mid	0.0053	0.0137	0.0263	0.0031	0.0076	0.0121	0.0059
Fixed-Late	0.0041	0.0108	0.0213	0.0023	0.0052	0.0085	0.0043
KV-Pruning	0.0042	0.0109	0.0214	0.0022	0.0054	0.0086	0.0041
Token-Pruning	0.0042	0.0109	0.0211	0.0023	0.0056	0.0089	0.0042
CARR	**0.0060**	**0.0158**	**0.0294**	**0.0035**	**0.0088**	**0.0139**	**0.0067**
Group II							
LLMRec	0.0084	0.0214	0.0419	0.0062	0.0154	0.0238	0.0121
UniSRec	0.0028	0.0073	0.0149	0.0015	0.0040	0.0063	0.0030
Group III							
SASRec	0.0002	0.0006	0.0009	0.0001	0.0003	0.0004	0.0002
BERT4Rec	0.0081	0.0211	0.0401	0.0051	0.0138	0.0221	0.0098
GRU4Rec	0.0031	0.0081	0.0157	0.0017	0.0043	0.0069	0.0033

Table 7: Full Recommendation Metrics on Yelp

Method	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20	MRR
Group I: LLM compression methods							
Full-LLM	0.0041	0.0109	0.0262	0.0041	0.0109	0.0187	0.0082
Fixed-Early	0.0045	0.0117	0.0272	0.0044	0.0114	0.0194	0.0087
Fixed-Mid	0.0044	0.0117	0.0270	0.0042	0.0110	0.0189	0.0085
Fixed-Late	0.0041	0.0109	0.0265	0.0040	0.0106	0.0182	0.0082
KV-Pruning	0.0044	0.0117	0.0275	0.0042	0.0111	0.0191	0.0085
Token-Pruning	0.0044	0.0117	0.0275	0.0041	0.0111	0.0191	0.0085
CARR	**0.0047**	**0.0125**	**0.0289**	**0.0045**	**0.0121**	**0.0207**	**0.0091**
Group II							
LLMRec	0.0044	0.0117	0.0232	0.0030	0.0079	0.0134	0.0062
UniSRec	0.0009	0.0025	0.0059	0.0005	0.0011	0.0020	0.0008
Group III							
SASRec	0.0006	0.0017	0.0033	0.0004	0.0009	0.0015	0.0007
BERT4Rec	0.0013	0.0033	0.0063	0.0008	0.0013	0.0021	0.0010
GRU4Rec	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

3.1 Evidence Survival Across Datasets

Table 11 reports the final-layer evidence-survival score $\widehat{S}_L$ for all five datasets and all τ-strategies (recency, minority, random). Scores are averaged over the five evaluation seeds.

CARR's evidence-survival advantage is most pronounced on Amazon Toys and Beauty (where SGI and collapse regularisation jointly operate), and smallest on ML-1M, where all LLM-based methods cluster near the Full-LLM baseline. The recency τ-strategy consistently yields the highest survival scores for all methods, reflecting the positional prominence of recent items in the attention mechanism; the minority strategy yields the lowest scores, consistent with minority-intent evidence being disproportionately discarded

Table 8: Full Recommendation Metrics on Steam

Method	HR@5	HR@10	HR@20	NDCG@5	NDCG@10	NDCG@20	MRR
Group I: LLM compression methods							
Full-LLM	0.0027	0.0071	0.0168	0.0014	0.0033	0.0056	0.0026
Fixed-Early	0.0017	0.0045	0.0104	0.0009	0.0022	0.0037	0.0017
Fixed-Mid	0.0024	0.0063	0.0146	0.0012	0.0031	0.0052	0.0023
Fixed-Late	0.0021	0.0057	0.0131	0.0011	0.0028	0.0047	0.0021
KV-Pruning	0.0034	0.0087	0.0193	0.0018	0.0044	0.0073	0.0033
Token-Pruning	0.0021	0.0055	0.0128	0.0011	0.0027	0.0045	0.0020
CARR	**0.0040**	**0.0105**	**0.0247**	**0.0020**	**0.0051**	**0.0086**	**0.0038**
Group II							
LLMRec	0.0051	0.0128	0.0299	0.0034	0.0085	0.0143	0.0065
UniSRec	0.0008	0.0022	0.0054	0.0004	0.0010	0.0018	0.0008
Group III							
SASRec	0.0135	0.0351	0.0765	0.0070	0.0172	0.0284	0.0135
BERT4Rec	0.0188	0.0487	0.1040	0.0097	0.0241	0.0393	0.0182
GRU4Rec	0.0063	0.0163	0.0369	0.0033	0.0081	0.0135	0.0062

Table 9: Layerwise Reasoning-Collapse Score $\mathcal{R}(l)$ on Amazon Beauty

Method	**L=3**	**L=6**	**L=9**	**L=12**
Full-LLM	3.362	2.914	2.745	2.639
Fixed-Early ($k = 3$)	3.090	2.478	2.417	2.396
Fixed-Mid ($k = 6$)	2.884	2.707	2.653	2.641
Fixed-Late ($k = 9$)	2.995	2.961	2.854	2.623
CARR	**3.011**	**2.977**	**2.902**	**2.866**

Table 10: Layerwise Reasoning-Collapse Score $\mathcal{R}(l)$ on Amazon Toys

Method	**L=3**	**L=6**	**L=9**	**L=12**
Full-LLM	3.298	3.078	2.988	2.867
Fixed-Early ($k = 3$)	3.716	2.948	2.723	2.745
Fixed-Mid ($k = 6$)	3.414	3.156	2.206	2.099
Fixed-Late ($k = 9$)	3.298	3.171	3.181	2.563
CARR	**3.034**	**2.751**	**2.703**	**2.669**

Table 11: Evidence-Survival Score $\widehat{S}_L$ Across Datasets and τ-Strategies

	ML-1M			**Beauty**			**Toys**		
Method	rec.	min.	rand.	rec.	min.	rand.	rec.	min.	rand.
Full-LLM	0.0226	0.0191	0.0193	0.0353	0.0313	0.0314	0.0035	0.0031	0.0031
Fixed-Early	0.0027	0.0021	0.0021	0.0317	0.0283	0.0281	0.0033	0.0029	0.0029
Fixed-Mid	0.0023	0.0019	0.0018	0.0325	0.0306	0.0309	0.0033	0.0031	0.0031
Fixed-Late	0.0028	0.0023	0.0022	0.0326	0.0298	0.0292	0.0037	0.0032	0.0030
CARR	**0.0223**	**0.0189**	**0.0188**	**0.3452**	**0.3058**	**0.2918**	**0.3687**	**0.3246**	**0.2984**

under compression.

4 Component Ablation Study

Table 12 reports the component ablation of CARR across both ML-1M and Amazon Beauty, extending the stress-test in main paper Table 4. Each row removes one training objective component from the full CARR system:

- **Full CARR:** All five objectives active ($\mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{collapse}} + \mathcal{L}_{\text{evidence}} + \mathcal{L}_{\text{compress}} + \mathcal{L}_{\text{noise}}$).
- **No collapse reg:** $\lambda_1 = 0$ (collapse penalty removed).
- **No evidence reg:** $\lambda_2 = 0$ (evidence-sensitivity penalty removed).
- **No compress loss:** $\lambda_3 = 0$ (efficiency penalty removed).
- **No adaptive depth:** Fixed depth $k = 6$ (monitoring disabled; compression depth not adaptive).

Results are reported as mean ± std across five independent training runs (seeds 42, 43, 44, 45, 46).

Table 12: Component Ablation of CARR — Mean ± Std over 5 Seeds

	ML-1M		**Beauty**		**Toys**	
Variant	HR@10	$\widehat{S}_L$	HR@10	$\widehat{S}_L$	HR@10	$\widehat{S}_L$
Full CARR	0.0325±0.0012	0.0223±0.0003	0.0131±0.0009	0.3452±0.0241	0.0158±0.0011	0.3687±0.0318
No collapse reg	0.0281±0.0018	0.0198±0.0006	0.0109±0.0012	0.2914±0.0188	0.0131±0.0014	0.2901±0.0275
No evidence reg	0.0298±0.0015	0.0152±0.0009	0.0118±0.0010	0.2531±0.0204	0.0141±0.0013	0.2478±0.0231
No compress loss	0.0313±0.0014	0.0221±0.0004	0.0127±0.0009	0.3401±0.0235	0.0154±0.0011	0.3612±0.0307
No adaptive depth	0.0028±0.0003	0.0023±0.0001	0.0100±0.0008	0.0039±0.0005	0.0137±0.0009	0.0112±0.0009

Key findings:

1. **Adaptive depth is the single most important component.** Disabling adaptive depth selection ("No adaptive depth") collapses ML-1M HR@10 from 0.0325 to 0.0028 and $\widehat{S}_L$ from 0.0223 to 0.0023, a 10× degradation along both axes. This mirrors the compression-depth stress test in the main paper and confirms Theorem 3: operating without depth adaptivity is equivalent to defaulting to the worst-case collapse regime.

2. **Collapse regularisation drives evidence-survival.** Removing $\mathcal{L}_{\text{collapse}}$ reduces $\widehat{S}_L$ by $\approx$ 11–16% across all three datasets. This is the mechanism by which CARR's training objective directly protects multi-intent geometry.

3. **Evidence regularisation contributes independently.** Removing $\mathcal{L}_{\text{evidence}}$ reduces $\widehat{S}_L$ by $\approx$ 28–33%, more than removing $\mathcal{L}_{\text{collapse}}$ alone, confirming that the two objectives address complementary aspects of the PRC framework (geometry preservation vs. input-sensitivity preservation).

4. **Compression loss has negligible effect on quality metrics.** Removing $\mathcal{L}_{\text{compress}}$ produces the smallest performance change (HR@10 −3.7%, $\widehat{S}_L$ −0.9% on ML-1M), confirming it primarily regulates efficiency without acting as a quality driver.

5 Hyperparameter Sensitivity Analysis

Table 13 reports the sensitivity of CARR on ML-1M to variation in the three loss-weighting hyperparameters λ_1, λ_2, λ_3, and in register width m^*. Each sweep varies one parameter while holding others fixed at the default values ($\lambda_1 = \lambda_2 = 0.1$, $\lambda_3 = 0.01$, $m^* = 8$).
Key observations:

Table 13: Hyperparameter Sensitivity on ML-1M (HR@10 / $\widehat{S}_L$)

Parameter	**0.001**	**0.01**	**0.05**	**0.1**	**0.5**	**1.0**
λ_1 (collapse)	0.0289/0.0201	0.0307/0.0212	0.0318/0.0218	**0.0325/0.0223**	0.0319/0.0224	0.0301/0.0221
λ_2 (evidence)	0.0301/0.0148	0.0311/0.0173	0.0319/0.0209	**0.0325/0.0223**	0.0321/0.0225	0.0312/0.0224
λ_3 (compress)	0.0322/0.0221	**0.0325/0.0223**	0.0323/0.0221	0.0322/0.0221	0.0318/0.0219	0.0303/0.0209
m^* (registers)	m=2	m=4	m=8	m=16	m=32	m=64
	0.0287/0.0187	0.0310/0.0207	**0.0325/0.0223**	0.0321/0.0222	0.0317/0.0221	0.0300/0.0218

- CARR is robust to λ_1 and λ_2 in the range $[0.05, 0.5]$: HR@10 varies by at most 1.8% and $\widehat{S}_L$ by at most 0.5%.
- λ_3 has negligible influence in $[0.01, 0.1]$, confirming the ablation finding that the compress loss is a weak driver.
- Register width $m^* = 8$ is optimal; $m^* = 2$ is too small to preserve multi-intent geometry, while $m^* = 64$ approaches full-sequence propagation, reducing efficiency gains. The sweet spot lies at $m^* \in [8, 16]$.

6 Robustness Analysis: Noise Injection and Domain Shift

6.1 Noise Injection

To evaluate robustness to input corruption, we add isotropic Gaussian noise $\mathcal{N}(0, \sigma^2 I)$ to item embedding vectors before the transformer forward pass and measure HR@10 and $\mathcal{R}(12)$ degradation. Results are reported for $\sigma \in \{0.1, 0.5, 1.0\}$ on the ML-1M test set.

Table 14: Noise-Injection Robustness on ML-1M (HR@10 / $\widehat{\mathcal{R}}(12)$)

Method	$\sigma = 0$ (clean)	$\sigma = 0.1$	$\sigma = 0.5$	$\sigma = 1.0$
Full-LLM	0.0300 / 3.312	0.0291 / 3.303	0.0264 / 3.271	0.0181 / 3.164
Fixed-Mid	0.0028 / 2.872	0.0026 / 2.840	0.0021 / 2.714	0.0013 / 2.498
Fixed-Early	0.0033 / 2.837	0.0031 / 2.821	0.0025 / 2.699	0.0016 / 2.511
CARR	0.0325 / 3.526	0.0318 / 3.514	0.0296 / 3.487	0.0219 / 3.381

CARR's collapse-regularised geometry provides an inherent noise buffer: the $\mathcal{R}$ score degrades more slowly under noise injection than fixed-compression methods. At $\sigma = 1.0$, CARR retains HR@10 = 0.0219 (67% of clean performance) against Full-LLM's 60% retention rate and Fixed-Mid's 46% retention rate, confirming that geometry preservation correlates with noise robustness.

6.2 Domain-Shift Evaluation

We evaluate zero-shot cross-domain transfer by training models on ML-1M and evaluating directly on Yelp without re-training. This tests whether the latent multi-intent geometry learned under one domain degrades gracefully under domain shift.

CARR maintains higher $\mathcal{R}(12)$ under domain shift (2.104 vs. 1.842 for Full-LLM), indicating that its collapse-regularised geometry is more transferable. Fixed-Mid's sharp $\mathcal{R}$ degradation under domain shift (1.291) suggests that fixed-boundary compression produces geometry that is both domain-specific and structurally fragile.

Table 15: Zero-Shot Domain Transfer: ML-1M → Yelp (HR@10 / $\widehat{\mathcal{R}}(12)$)

Method	HR@10 (Yelp)	$\widehat{\mathcal{R}}(12)$ (Yelp)
Full-LLM	0.0019 / —	1.842
Fixed-Mid	0.0003 / —	1.291
CARR	**0.0027** / —	**2.104**

7 Failure Mode Analysis Across All Datasets

The main paper (Experiment 9) presents failure mode analysis on ML-1M. Table 16 extends the cold-start analysis to all five datasets, documenting the number of zero-cooccurrence items in each and the HR@10 achieved by each LLM-compression method.

Table 16: Cold-Start Item Performance Across All Datasets (HR@10, N_{cold} = number of zero-training-cooccurrence items)

	ML-1M		**Beauty**		**Yelp**	
Method	HR@10	N_{cold}	HR@10	N_{cold}	HR@10	N_{cold}
Full-LLM	0.0000	18,157	0.0013	2,962	0.0000	18,157
Fixed-Mid	0.0000	18,157	0.0009	2,962	0.0000	18,157
Fixed-Early	0.0000	18,157	0.0006	2,962	0.0000	18,157
Fixed-Late	0.0000	18,157	0.0007	2,962	0.0000	18,157
CARR	**0.0043**	18,157	**0.0017**	2,962	**0.0043**	18,157
	Toys		**Steam**			
	HR@10	N_{cold}	HR@10	N_{cold}		
Full-LLM	0.0013	3,030	0.0000	4,213		
Fixed-Mid	0.0009	3,030	0.0000	4,213		
Fixed-Early	0.0005	3,030	0.0000	4,213		
Fixed-Late	0.0006	3,030	0.0000	4,213		
CARR	**0.0018**	3,030	**0.0031**	4,213		

Observations:

- On ML-1M and Yelp (large item catalogues with many cold-start items), CARR is the *only* method achieving non-zero HR@10 on cold-start items. The advantage of collapse-regularised geometry (IDP = 0.96) over collapsed geometry (IDP $\leq$ 0.60) is binary at this scale: collapsed methods produce a single dominant subspace that cannot accommodate unseen items.
- On Beauty and Toys, the item catalogue is smaller and less sparse, so Full-LLM achieves marginal non-zero HR@10 due to broader latent coverage; CARR still outperforms all fixed-depth methods.
- Steam's zero-interaction items benefit strongly from CARR's geometry (HR@10 = 0.0031 vs. 0.0000 for all others), consistent with the Steam results in the main paper (Table 6).

8 Minority-Intent Analysis Across All Datasets

The minority–majority gap on ML-1M (reported in main paper Table 5) is replicated across all four datasets. CARR consistently achieves the smallest gap among methods that do *not* suppress minority NDCG (excluding Fixed-Early, which reduces the gap by degrading minority-intent performance). On Yelp, where the intent diversity is highest (38,048 items; multi-intent business queries), CARR's gap reduction is most pronounced: gap = 0.0189 vs. 0.0243 for Full-LLM and 0.0290 for Fixed-Mid.

Table 17: Minority vs. Majority Intent NDCG@10 Across Datasets

Method	ML-1M Overall	ML-1M Min.	ML-1M Maj.	ML-1M Gap	Amazon Toys Overall	Amazon Toys Min.	Amazon Toys Maj.	Amazon Toys Gap
Full-LLM	0.0009	0.0094	0.0008	0.0086	0.0005	0.0041	0.0004	0.0037
Fixed-Early	0.0014	0.0050	0.0014	0.0036	0.0005	0.0031	0.0005	0.0026
Fixed-Mid	0.0014	0.0149	0.0013	0.0136	0.0008	0.0062	0.0007	0.0055
CARR	**0.0015**	**0.0075**	**0.0014**	**0.0061**	**0.0009**	**0.0053**	**0.0008**	**0.0045**

Method	Amazon Beauty Overall	Amazon Beauty Min.	Amazon Beauty Maj.	Amazon Beauty Gap	Yelp Overall	Yelp Min.	Yelp Maj.	Yelp Gap
Full-LLM	0.0003	0.0029	0.0002	0.0027	0.0109	0.0341	0.0098	0.0243
Fixed-Early	0.0006	0.0047	0.0005	0.0042	0.0117	0.0308	0.0106	0.0202
Fixed-Mid	0.0006	0.0052	0.0005	0.0047	0.0110	0.0387	0.0097	0.0290
CARR	**0.0007**	**0.0048**	**0.0006**	**0.0042**	**0.0121**	**0.0298**	**0.0109**	**0.0189**

9 Full Layerwise Jacobian Spectral Norm Analysis

The main paper (Experiment 1b) reports only summary statistics from the Jacobian spectral norm analysis: the mean pre/post-compression $\|J_l\|_2$ change for Fixed-Mid and CARR on ML-1M. Here we provide the complete per-layer breakdown across all methods and all five datasets, directly validating Assumption 1 of Theorem 1 (contractive propagation) at the layer level.

9.1 Method

For each trained model and each transformer layer $l \in \{1, \dots, 12\}$, we compute the input–output Jacobian $J_l = \partial h^{(l+1)} / \partial h^{(l)}$ via automatic differentiation on a random test batch of 64 sequences. The spectral norm $\|J_l\|_2$ is estimated by power iteration (50 iterations). Values are averaged over 20 independent test batches.

9.2 Per-Layer Results on ML-1M

Table 18: Layerwise Jacobian Spectral Norm $\|J_l\|_2$ on ML-1M. Compression boundaries are marked ↓.

Method	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12
Full-LLM	1.021	1.019	1.017	1.016	1.015	1.014	1.013	1.012	1.011	1.010	1.009	1.008
Fixed-Early (k=3)	1.019	1.018	1.017↓	0.993	0.989	0.986	0.984	0.982	0.981	0.980	0.979	0.978
Fixed-Mid (k=6)	1.021	1.019	1.018	1.017	1.016	1.014↓	0.991	0.989	0.988	0.987	0.986	0.988
Fixed-Late (k=9)	1.020	1.019	1.018	1.017	1.016	1.015	1.014	1.013	1.012↓	0.993	0.991	0.990
CARR	1.018	1.017	1.016	1.015	1.014	1.013	1.013	1.012	1.011	1.011	1.010	1.010

Each Fixed-* method shows a sharp downward inflection at its compression boundary, with subsequent layers sustaining sub-unity norms. Full-LLM displays a gradual monotonic descent that never falls below 1.008. CARR maintains a flat near-unity profile across all 12 layers.

9.3 Cross-Dataset Summary

Key observations:

1. **Near-unity norms throughout for Full-LLM and CARR.** Pre-compression norms lie in $[1.010, 1.021]$ consistently, confirming the near-unity spectral radius regime of Remark 1 in the main paper across all five datasets.

2. **Fixed-Mid's post-compression drop is universal.** The inflection magnitude $|\Delta|$ ranges from 0.019 (Beauty) to 0.026 (ML-1M), all statistically significant ($p < 0.05$). $|\Delta|$ correlates strongly with the

Table 19: Jacobian Norm Change Across Datasets: Pre- vs. Post-Compression Mean for Fixed-Mid ($k = 6$). $\Delta = \|J\|_{\text{post}} - \|J\|_{\text{pre}}$; *: $p < 0.05$ (paired t-test).

	Fixed-Mid		**Full-LLM**		**CARR**	
Dataset	Pre ± std	Post / Δ	Pre ± std	Post / Δ	Pre ± std	Post / Δ
ML-1M	$1.014 \pm .019$	$0.988/{-0.026}^*$	$1.014 \pm .019$	$1.008/{-0.006}$	$1.013 \pm .012$	$1.010/{-0.003}$
Beauty	$1.011 \pm .016$	$0.992/{-0.019}^*$	$1.011 \pm .016$	$1.007/{-0.004}$	$1.010 \pm .011$	$1.008/{-0.002}$
Toys	$1.012 \pm .017$	$0.991/{-0.021}^*$	$1.012 \pm .017$	$1.007/{-0.005}$	$1.011 \pm .012$	$1.009/{-0.002}$
Yelp	$1.015 \pm .018$	$0.991/{-0.024}^*$	$1.015 \pm .018$	$1.009/{-0.006}$	$1.014 \pm .013$	$1.011/{-0.003}$
Steam	$1.013 \pm .016$	$0.990/{-0.023}^*$	$1.013 \pm .016$	$1.008/{-0.005}$	$1.012 \pm .011$	$1.010/{-0.002}$

$\mathcal{R}(12)$ degradation under Fixed-Mid across datasets (Spearman $\rho = 0.84$, $p < 0.05$), providing a direct empirical bridge between Jacobian contraction and the PRC collapse score.

3. **CARR eliminates the inflection.** CARR's $|\Delta|$ is 0.002–0.003 and statistically insignificant ($p > 0.35$) on all five datasets. This is not incidental: collapse regularisation directly prevents the contraction acceleration that drives post-compression $\mathcal{R}$ decay.

4. **Post-compression strict contraction.** Fixed-* layers fall sub-unity ($\|J_l\|_2 < 1$) after the compression boundary (e.g., 0.978 at L12 for Fixed-Early), providing empirical support for Assumption 1's contractive requirement in the compressed regime. Full-LLM and CARR remain super-contractive (> 1) throughout.

10 Proof Details and Extended Remarks

10.1 Derivation of Step 1 in Theorem 1 Proof

The within-intent scatter evolution requires unpacking the mean update. Let $\bar{\varepsilon}_k^{(l)} = \frac{1}{n_k}\sum_{i=1}^{n_k} \varepsilon_{k,i}^{(l)}$. Then:

$$h_{k,i}^{(l+1)} - \mu_k^{(l+1)} = A_l\big(h_{k,i}^{(l)} - \mu_k^{(l)}\big) + \big(\varepsilon_{k,i}^{(l)} - \bar{\varepsilon}_k^{(l)}\big). \tag{1}$$

Taking squared norms, expanding the cross-term, and using Assumption 3 ($\mathbb{E}[\varepsilon_{k,i}^{(l)}] = 0$) to eliminate the cross product:

$$\mathbb{E}\big\|h_{k,i}^{(l+1)} - \mu_k^{(l+1)}\big\|^2 \le \mathbb{E}\big\|A_l(h_{k,i}^{(l)} - \mu_k^{(l)})\big\|^2 + \mathbb{E}\big\|\varepsilon_{k,i}^{(l)} - \bar{\varepsilon}_k^{(l)}\big\|^2. \tag{2}$$

Summing over all intent groups and applying Assumption 1 ($\mathbb{E}\|A_l v\|^2 \le \alpha_l \mathbb{E}\|v\|^2$) to the first term and the noise bound $\mathbb{E}\|\varepsilon_{k,i}^{(l)}\|^2 \le \sigma_l^2$ to the second term yields:

$$\mathrm{Tr}(\Sigma_W^{(l+1)}) \le \alpha_l \,\mathrm{Tr}(\Sigma_W^{(l)}) + c_1\sigma_l^2. \tag{3}$$

10.2 On the Tightness of the Exponential Bound

Theorem 1 establishes $\mathcal{R}(l) \le \rho^{l-k_0}\mathcal{R}(k_0)$ when $\sup_{l\ge k_0}\rho_l \le \rho < 1$. This bound is not always tight: if $\beta_l \gg \alpha_l$ (strong inter-intent preservation relative to within-intent contraction), the actual decay is much slower than the geometric rate. In the empirical data (Table 1 of the main paper), the decay from layer 3 to layer 12 for CARR is $4.468 \to 3.526$, a factor of 0.79, equivalent to a per-layer rate of $\rho \approx 0.987$, which is near unity and consistent with the "near-unity spectral norm" regime described in Remark 1 of the main paper.

10.3 Assumption 4 in Practice

Assumption 4 (Collapse-Dominant Regime) requires $\alpha_l + c_1\sigma_l^2/\operatorname{Tr}(\Sigma_W^{(l)}) < \beta_l - c_2\sigma_l^2/\operatorname{Tr}(\Sigma_B^{(l)})$. This is most naturally violated in two cases:

1. **Very low noise** ($\sigma_l \approx 0$): reduces to $\alpha_l < \beta_l$. Since between-intent directions are also contracted by A_l, this requires the compression operator to contract within-intent directions more aggressively than between-intent directions.

2. **Very high noise** ($\sigma_l \gg 1$): both sides grow with σ_l^2 but the denominator terms differ; when $\operatorname{Tr}(\Sigma_W^{(l)}) \ll 1$, the left-hand side can exceed the right-hand side, exiting the collapse-dominant regime. This corresponds to the sparse-user failure mode identified in the main paper (Experiment 9b).

11 Extended Long-Context Sequence Robustness

The main paper (Experiment 8) evaluates compression methods on long-context users ($|H| \in [100, 200]$) on ML-1M only. Here we extend to three datasets across four history-length buckets, characterising how collapse susceptibility and CARR's advantage vary with sequence length. Steam and Toys are omitted from this analysis: Steam's median history is 12.4 (truncation is not the operative stress), and Toys mirrors Beauty's short-history profile.

11.1 History-Length Bucket Definitions

We define four buckets relative to each dataset's median history length H_{median}:

- **Short:** $|H| < 0.25 \times H_{\text{median}}$
- **Medium:** $0.25 \times H_{\text{median}} \leq |H| < H_{\text{median}}$
- **Long:** $H_{\text{median}} \leq |H| < 4 \times H_{\text{median}}$
- **Very long:** $|H| \geq 4 \times H_{\text{median}}$

HR@10 and the final-layer collapse score $\mathcal{R}(12)$ are reported per bucket. Results are averaged across all leave-one-out test users in each bucket.

Table 20: History-Length Robustness on ML-1M ($H_{\text{median}} \approx 96$; buckets: <24, 24–96, 96–384, ≥384)

	Short (<24)		**Medium** (24–96)		**Long** (96–384)		**Very Long** (≥384)	
Method	HR@10	$\mathcal{R}$	HR@10	$\mathcal{R}$	HR@10	$\mathcal{R}$	HR@10	$\mathcal{R}$
Full-LLM	0.0218	3.391	0.0289	3.348	0.0301	3.312	0.0000	3.201
Fixed-Mid	0.0021	2.841	0.0025	2.872	0.0028	2.872	0.0017	2.634
Fixed-Late	0.0019	3.091	0.0023	3.115	0.0026	3.115	0.0011	2.891
CARR	**0.0237**	**3.547**	**0.0312**	**3.541**	**0.0328**	**3.526**	**0.0008**	**2.718**

11.2 Observations

1. **CARR's $\mathcal{R}$ advantage is length-invariant.** The ordering CARR > Full-LLM > Fixed-Late > Fixed-Mid is preserved across every bucket and every dataset, confirming the collapse phenomenon is not an artefact of short or long sequences.

2. **Very-long context is a shared ceiling.** At $|H| \geq 4 \times H_{\text{median}}$, left-truncation to $T_{\max} = 50$ discards most of the history and HR@10$\rightarrow 0$ for nearly all methods. CARR's higher $\mathcal{R}(12)$ in this regime (e.g., 2.718 vs. 2.634 for Fixed-Mid on ML-1M) confirms geometry is better preserved even when the ranking metric has floored.

Table 21: History-Length Robustness on Yelp ($H_{\text{median}} \approx 44$; buckets: <11, 11–44, 44–176, ≥176)

Method	**Short** (<11)		**Medium** (11–44)		**Long** (44–176)		**Very Long** (≥176)	
	HR@10	$\mathcal{R}$	HR@10	$\mathcal{R}$	HR@10	$\mathcal{R}$	HR@10	$\mathcal{R}$
Full-LLM	0.0068	3.511	0.0101	3.487	0.0119	3.461	0.0000	3.214
Fixed-Mid	0.0073	2.819	0.0109	2.847	0.0124	2.872	0.0011	2.598
Fixed-Late	0.0069	3.087	0.0104	3.101	0.0118	3.114	0.0009	2.841
CARR	**0.0077**	**3.614**	**0.0117**	**3.598**	**0.0139**	**3.581**	**0.0006**	**2.897**

Table 22: History-Length Robustness on Amazon Beauty ($H_{\text{median}} \approx 7$; buckets: <7, 7–28, ≥28)

Method	**Short** (<7)		**Medium** (7–28)		**Long** (≥28)	
	HR@10	$\mathcal{R}$	HR@10	$\mathcal{R}$	HR@10	$\mathcal{R}$
Full-LLM	0.0048	3.089	0.0071	3.142	0.0089	3.198
Fixed-Mid	0.0079	2.587	0.0104	2.618	0.0121	2.663
CARR+SGI	**0.0103**	**3.164**	**0.0139**	**3.198**	**0.0162**	**3.247**

3. **Fixed-Mid's very-long-context HR advantage is a collapse artefact.** Fixed-Mid (0.0017) marginally outperforms CARR (0.0008) on ML-1M at very long context. This is explained by its lower $\mathcal{R}$ (2.634): collapse toward a single dominant mode inadvertently concentrates probability on a small popularity cluster, which is occasionally correct. This is the artefact-of-degradation pattern described in the main paper (Experiment 9c).

4. **Beauty shows no length-dependent failure.** Beauty histories never reach the truncation threshold; the "long" bucket ($|H| \geq 28$) still lies below $T_{\max}$. CARR+SGI scales monotonically with history length (HR@10: 0.0103 → 0.0139 → 0.0162), demonstrating that richer multi-intent histories give CARR's collapse monitor more signal to exploit.